No Detectable Self-Improvement from Generate–Verify–Train Iterations: A Negative Result with Qwen2.5-1.5B, LoRA SFT and a Rule-Based Verifier

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Abstract. Generate → verify → select → train (GVT) self-training is said to let a small language model improve its own reasoning without new human labels. We test that claim on Qwen2.5-1.5B-Instruct with LoRA fine-tuning and the rule-based verifier `math-verify`, which rejects some correct answers because of how they are written. Our question is whether the model gets better at mathematics or only learns to write what the verifier accepts. We fixed four measures in advance (pass@1, pass@k, output style and per-problem gains and losses), read on a held-out exam of 150 problems; measures of termination and of verifier error were added after the runs. We find no detectable improvement. On MATH-500 at four learning rates from 10^{-5} to 5×10^{-4} (two or three seeds each, three controls) and on GSM8K at two rates (one seed each), no pass@1 or pass@k gain has a problem-level bootstrap interval excluding zero (their upper ends reach +8.0 and +7.3 points), and the intervals for sample accuracy, which we chose as the primary measure after the runs, exclude gains above 2.2 points in every run. These intervals omit training-seed variance (a replicate of one configuration ends 3.2 points of sample accuracy away) and are not a pre-specified equivalence test, so they bound the effect only loosely. Whether the model learns the verifier's style is not established either way: our pre-set rule cannot detect it while accuracy falls, and the `\boxed` share does not discriminate. Accuracy falls in every run at the highest rate; that the decline grows with the rate rests on one seed. At 5×10^{-4}, generations that finish within the 512-token cap are 5–11 points less accurate than the untrained model's on the same problems, and the share that hits the cap, already 43% for the untrained model, rises 2–14 points in six runs (not in a replicate of one of them); by run, 34–74% of the sample-accuracy decline lies in finished accuracy and the rest in termination. At lower rates the cap-hit share does not rise (changes of −3.3 to +1.1 points). A further verifier defect (bare-stored gold against boxed predictions) rejects 15.5–18.6% of the loop's correct solutions and lowers absolute pass@1 by about 5–8 points; re-scored without it, sample accuracy still falls in all five runs we re-scored. Patching a rarer defect and removing the filter make no consistent difference, but both controls are underpowered and differ in training dose. A sharp decline in an earlier pooled design does not repro-

duce under isolation and held-out scoring, but that design also trained on about 4.5× more examples per round, so its cause is not identified.

Keywords: self-training · self-improvement · verifier bias · mathematical reasoning · small language models · negative result.

1 Introduction

A common recipe for improving the mathematical reasoning of a small language model (SLM) *without human annotation* is to let the model generate its own solutions, keep those that an automated verifier judges correct, and fine-tune on the kept solutions, for several rounds. We call this the *Generate–Verify–Train* (GVT) loop; it underlies STaR [16], ReST [10], ReST-EM [9] and rejection-sampling fine-tuning [14]. The usual reading is that the model "self-improves". But the verifier never reads the reasoning: a rule-based verifier such as `math-verify`[1] only extracts a final answer and compares it with the gold answer, and it rejects some correct answers written in forms it does not parse (Section 3.2). The training signal is therefore "was the final string accepted", not "were the steps right".

This paper asks one question: *after several GVT rounds, is the model better at mathematics, or has it only learned to write what the verifier accepts?* We fixed four measures before running the loop. (i) pass@1; (ii) pass@k, the share of problems solved at least once in k samples; (iii) output style, chiefly the share of generations containing `\boxed`, the delimiter the verifier reads; and (iv) per-problem transitions between rounds: problems newly solved (*added*) and problems no longer solved (*lost*). We set the reading rule in advance: "learning the verifier's style" requires pass@1 to rise, pass@k to stay flat and style to converge, all at once; any missing piece means "not established". In hindsight the rule is one-sided: it can only fire when accuracy rises, so it cannot detect verifier-style learning that happens alongside a general decline (Section 5.1). Measures of termination (cap-hit share, accuracy on finished generations) and the re-scoring for a second verifier defect were added after the runs, in response to what the pre-set measures showed; we mark them as post hoc where they carry a claim.

Our answer is negative on the first branch and open on the second. We detect no improvement on held-out problems at any learning rate we tried, and the pattern our rule looks for does not appear, although for the reason just given that does not show the model ignores the verifier. What does appear, most clearly at the highest rate, is lower accuracy on the solutions the model does finish and, at that rate, more generations that run out of tokens. We reached this answer through two designs whose early readings did not hold up, and we report them because the reasons are instructive: a pooled design whose sharp decline did not reproduce under isolation (Section 4.1), and an isolated design at a learning rate that changed the outputs little (Section 4.2).

[1] https://github.com/huggingface/math-verify, version 0.9.0 with antlr4 4.13.2 in this work.

Contributions.

- A direct answer, for a 1.5B model, LoRA SFT and one rule-based verifier, to whether GVT self-training improves mathematical reasoning: no gain is detectable on a held-out exam at any of four learning rates on MATH-500 or two on GSM8K; sample-accuracy intervals (sample accuracy being the primary measure chosen after the runs) exclude gains above 2.2 points in every run, with problem-level bootstrap intervals for every comparison. These intervals omit training-seed variance and are not an equivalence test, so the bound is loose.
- An observation on style, with its limits: the `\boxed` share falls under self-training, but among finished generations it starts above the verifier-selected pool and falls towards it, so it cannot tell imitation of the pool from drift; the only strong style convergence occurs when training on reference solutions, where no verifier is in the loop and accuracy falls. Our pre-set rule cannot rule out verifier-style learning.
- A post hoc description of what training at the highest rate changes, using the token count of each generation: at 5×10^{-4} on MATH-500 the accuracy of finished generations, compared with the untrained model on the same problems, falls 5.5–10.6 points, and this finished-accuracy part is 34–74% of the sample-accuracy decline depending on the run; the cap-hit share rises only at that rate, by 2–14 points in six runs (at lower rates it changes by -3.3 to $+1.1$). At lower rates finished accuracy is also lower, but the intervals mostly include zero. A character-length heuristic we first used overstated the role of unfinished generations, and we show why.
- Design lessons, stated with their limits: a verifier-patch control whose treatment changes 0–4 training examples cannot detect an effect, which a prevalence count before the run would have shown, while a second, untested defect in the same pipeline affects about one in six of the loop's correct solutions; and a pooled, selection-scored design and an isolated design differ in training dose as well as in the variables we intended to vary.

2 Related Work

Self-training and its limits. STaR [16] introduced the generate–filter–retrain loop, and ReST [10], ReST-EM [9] and RFT [14] formalise it; ReST-EM, which like us restarts fine-tuning from the base model each iteration, reports diminishing test gains after the first round and a one-round regression on APPS. ReST filters machine-translation samples with a learned reward model and a rising threshold; the others filter by answer correctness. None of them measures how the filter's errors depend on the form of the answer, and they mostly use larger models than ours. Yue et al. [15] show that RL raises pass@1 while the base model wins at large pass@k, and Teacher-Free Self-Training [11] observes a pass@8 versus pass@64 crossover on a free-verifier domain, concluding that self-training "amplifies but does not compound". Training repeatedly on model-generated data erodes distribution tails [8], and Qiao et al. [7] argue theoretically

that a selector seeing only a narrow slice of the target distribution accelerates collapse.

Verifier error. TinyV [13] measures that rule-based verifiers reject many correct solutions and repairs them with an LLM check. From Accuracy to Robustness [5] reports an average recall of 86% for rule-based verifiers, i.e. about 14% of correct responses rejected, but for `math-verify` itself on their MATH subset a recall of 0.95 (about 5% false negatives; their Table 5, on DeepSeek-R1-Distill-Qwen-32B outputs). Our MATH-500 audit gives 7.8% (95% Wilson interval 4.3–13.8%) on another model's solutions, a figure biased low by its correctness check (Section 3.2), and a pipeline-specific defect alone rejects 15.5–18.6% of the loop's own correct solutions. These figures differ in verifier version, solution source and denominator, so we compare them only in order of magnitude. Imperfect Verifiers [1] gives the most formal noise model, with fixed false-accept and false-reject probabilities, and notes that real noise can depend on formatting but then assumes content-independent rates. Egashira et al. [3] classify verifier errors in RLVR on a synthetic arithmetic task (Qwen3-1.7B-Base, OLMo3-7B) with hand-designed patterns such as format-based false negatives, and find that systematic false negatives behave like random noise whereas false positives can cause plateau or collapse. Our setting differs in paradigm (offline rejection-sampling SFT with a fresh adapter per round, not on-policy RL) and in task (real benchmark problems). Our patch control cannot test their false-negative finding, because it changes too little of the training pool (Section 4.3), and the much larger false-negative class we measured afterwards is present in both arms. Nor does our no-filter control test their false-positive finding: their false positives are systematic (a fixed wrong pattern that is always accepted), whereas no-filter admits every sample, about two thirds of them incorrect in unstructured ways, under SFT with a 3–4× larger training dose. It did not end consistently worse than the filtered run, which we report only as a related observation.

Auditing self-improvement. Xu et al. [12] audit three rounds of rank-32 LoRA self-training of Qwen3-8B ($\eta = 10^{-4}$; filters by answer match, majority vote or an external teacher) against a frozen untrained control, and identify seven measurement failures that each invert a reported finding when its control is absent, including a fixed token cap under style drift and training-seed variance. Self-training does not improve rarely-solved problems and corrupts problems solved at baseline. This is the study closest to ours in spirit. Our per-problem added/lost counts point the same way (more problems lost than gained), though without their frozen control and with net counts that only restate the change in pass@k (Section 3.4), and our adapters also drift in style (longer outputs, fewer `\boxed` answers), and a fixed token cap matters in our setting too: the untrained model already hits it in 43% of MATH-500 generations. Their setup differs in model size, rank and filters; we lack their per-round frozen arm (we rescore the untrained model once, Section 4.6) and a distillation arm, and we add a measured format-biased verifier, a learning-rate series and reference-solution and no-filter controls.

Reference-guided training. Kim et al. [6] show that self-distillation conditioned on the ground-truth solution can cut mathematical accuracy by up to 40% (Qwen3-1.7B/8B, DeepSeek-Distill-Qwen-7B and Olmo3-7B-Instruct) while *shortening* responses, which they trace to suppressed epistemic verbalization. Our train-on-gold control is plain SFT on human reference solutions, not self-distillation, so it is at most a loose analogue; it shows a similar surface signature (shorter responses, lower accuracy) that may simply come from imitating terse references. We did not measure epistemic markers.

Positioning. In the literature we surveyed (searched through October 2026) we found no multi-round self-training study that uses a rule-based verifier with a measured format bias together with a matched patched-verifier control; the closest are Egashira et al. and Xu et al. Given that the patch turned out to be too weak to test anything (Section 4.3), we do not present this combination as the contribution. The contribution is the absence of a detectable gain under four pre-set measures with intervals, the description of what training at higher rates changes, and the design lessons.

3 Setup

3.1 The GVT Loop

Each round t has four steps. *Generate:* the current model samples k solutions per problem. *Verify:* `math-verify` extracts the final answer and compares it with the gold. *Select:* accepted solutions are kept. *Train:* a *fresh* LoRA adapter is attached to the *base* model and trained on the kept solutions for one epoch. Round 0 is the untrained base model; the round-t model ($t \geq 1$) is the base plus an adapter trained on the pool accepted at round $t-1$. Adapters are not accumulated, so earlier rounds affect later ones only through the data they generate; conclusions about (non-)compounding below apply to this restart-from-base design, which gives compounding less room than continued training would.

3.2 The Verifier and Its Directional Bias

Format probe. We re-rendered gold answers from GSM8K and MATH-500 in individually valid formats (bare number, trailing-zero decimal, `\frac`, `\dfrac`, `\tfrac`, boxed, dollar-delimited, sentence, with units) and asked the verifier to grade them against the original gold. On GSM8K plain rewrites are never rejected, while `\dfrac`/`\tfrac` are rejected 200/200 (the command is not parsed). On MATH-500, `\dfrac`/`\tfrac` are rejected 146/146 (100%), trailing-zero decimals 13/22, and boxed forms 41/200. Of those 41, 24 are gold answers our pipeline stores in forms the verifier cannot parse at all, so part of this figure measures our gold storage, not the verifier. The other 17 (17 of the 200 boxed probes, 8.5%) are compound answers such as $3\sqrt{13}$: the gold is stored without a delimiter, and `math-verify` applies its full LaTeX extraction only inside

`\boxed{}` or `$...$`, so the bare gold is parsed to its first number and no longer matches a correct boxed prediction.

False-negative rate on real solutions. We took one public solution per problem for the first 200 problems of each dataset (teacher traces for GSM8K, Qwen2.5-7B-Instruct rollouts for MATH-500), not the 1.5B model's own outputs. Correctness was established by a scripted comparison of the boxed answer with the gold string (`src/dacntt/read.py`), and we then counted how many of these correct solutions the verifier rejects. The false-negative rate is 7.8% on MATH-500 (10 of 128 correct solutions; 24 problems with unparseable gold excluded) and 0% on GSM8K (0/200; 95% Wilson interval 0–1.9%, one-sided upper bound 1.3%), identical for the `default` and `reward` extraction presets of `math-verify`. Of the ten MATH-500 cases, eight are wrong-token extraction from a long solution and two are valid rewrites the verifier cannot match. Two features of the audit bias the 7.8% low. Because correctness was a string comparison with the gold, a correct answer written in an equivalent but different form (e.g. `\frac43` for `\frac{4}{3}`) was classed as incorrect and left out of the denominator; that is exactly the class the gold-storage defect below rejects. And the 24 excluded problems are ones the verifier can never score correct. The audit also inspected only rejected solutions, so it cannot measure the false-positive rate; that the verifier rarely accepts wrong answers is an assumption here, not a measurement.

Unparseable gold in the loop. Over all 500 MATH-500 problems the verifier cannot parse 51 stored gold answers (10.2%); 11–18 of them fall in each 150-problem exam partition. The loop keeps these problems: every sample is rejected, so they never enter the training pool and always count as unsolved. They lower absolute pass rates by up to 12 points but are constant within a partition, so they cancel in within-run and same-partition comparisons.

The patch. For the control we repaired the most *severe* identified failure mode with a normalisation pass that rewrites `\dfrac` and `\tfrac` to `\frac` before parsing, removing the 100%-rejection class and nothing else. The patched verifier is therefore still biased: it keeps the wrong-token extractions (8 of the 10 audited false negatives) and the gold-storage defect.

The gold-storage defect in the loop's own outputs (post hoc). After the runs we re-scored the saved generations with a second normalisation that wraps the gold in `\boxed{}` when the prediction contains one and the gold does not (`experiments/viec3/review_cpu/`: `patch_dose.py`, `dose.py`, `flips.py`). This defect is far more frequent than the `\dfrac` class. Over all 500 problems it rejects 15.5–18.6% of the generations the re-scored verifier accepts (Run A at 5×10^{-4}, seeds 1 and 2, and at 10^{-5}, seed 0, every round), and accepting them would enlarge the kept pool by 17–23% (round 0 of Run A at seed 1: 440 → 542; at 10^{-5}: 872 → 1,066), against 0–4 examples (under 1%) for the `\dfrac` patch. In a hand check of 30 random flips, 29 pairs were the same value written differently (e.g. `\frac43` and `\frac{4}{3}`; one prediction had an empty box). The 15.5–18.6% assumes that the re-scored verifier itself accepts no wrong answer, which we did not measure. On the exam set the defect lowers absolute pass@1 by 6.7–8.0 points at round 0 and throughout the 10^{-5} runs, by 4.7 points at round 2

Table 1. The two designs. They differ in more than the intended variables; the training dose per round is the largest unintended difference.

	Design P (pooled)	Design I (isolated)
Datasets per run	GSM8K + MATH-500	one only
Adapter	one, shared	one per dataset
Training mixture	$\approx$69% GSM8K	single-domain
Evaluation items	same as selection	held-out 30%
Training examples, round 0 (1.5B)	3,962	872 (A), 1,901 (C)
Verifier control	none	Run B (patched)
Models	0.5B and 1.5B	1.5B

under 5×10^{-4}, and by 0.7–3.3 points for train-on-gold after round 0. The offset is therefore not constant: it is largest at round 0 and smaller after strong training, so with the original verifier the declines below are, if anything, *understated*. We keep the original verifier's numbers in every table, since that is the verifier the loop trained with; they are therefore about 5–8 points too low in absolute terms for self-training. The declines do not depend on it. Re-scored, with paired bootstrap intervals as in Section 3.4, sample accuracy changes by -10.7 [-14.8, -6.7] and -6.0 [-9.7, -2.2] points for Run A at 5×10^{-4} (seeds 1 and 2; original verifier -8.3 and -4.7), by -3.3 [-5.9, -0.7] for Run A at 10^{-5}, seed 0 (original -2.4), and by -24.7 and -22.0 for train-on-gold. In pass@1 the same runs go from 40.0% to 28.0% and from 38.0% to 32.0% (interval $[-14.0, +2.0]$ for the latter), from 40.0% to 36.7%, and from 40.0% to 16.0% and from 38.0% to 13.3%. Only these five runs were re-scored. Both arms of the patch control (Section 4.3) carry this defect.

3.3 Two Experimental Designs

Design P (pooled) loads both datasets (500 problems each) together and trains *one* adapter per round on the union of accepted solutions, with selection and evaluation on the same 1,000 problems. *Design I (isolated)* uses one dataset and one adapter per run, and holds out 30% of the 500 problems (150) as an `exam` set; generation runs on all 500 problems each round, but only the other 350 can enter the training pool, and all Design I pass rates are on the exam set. Table 1 summarises them.

Design I has three runs, all Qwen2.5-1.5B-Instruct, five rounds, $k = 8$, differing in one respect each: **Run A** (MATH-500, original verifier), **Run B** (MATH-500, patched verifier) and **Run C** (GSM8K, original verifier). Runs A and B were repeated at a second seed; since the seed also sets the train/exam partition, the repetition draws a different 150-problem exam set.

3.4 Models, Data, Training, Metrics

We use Qwen2.5-0.5B-Instruct and Qwen2.5-1.5B-Instruct (Design P uses both, Design I the 1.5B model only) and the first 500 problems of the GSM8K test

split [2] and MATH-500, the 500-problem subset of MATH [4]. The prompt is a single user turn in the model's chat template: "Solve the math problem. Put the final answer in `\boxed{}`." followed by the problem. Generation samples at temperature 0.8 with a cap of 512 new tokens; the other decoding settings come from the checkpoint's generation configuration (top-p 0.8, top-k 20, repetition penalty 1.1). Training is LoRA SFT (`transformers` 5.0.0, `peft` 0.19.1, TRL 1.10.0 `SFTTrainer`; rank 16, $\alpha = 32$, on `q_proj` and `v_proj`, LoRA dropout and weight decay at the library defaults; batch size 2; one epoch; AdamW with linear learning-rate decay and no warmup; loss over the whole chat sequence, prompt included; fp16 on one Kaggle T4), with a training sequence limit of 512 tokens that includes the prompt; all kept solutions are used, so the number of optimizer steps per round is half the pool size. The learning rate is 10^{-5} unless stated otherwise; this value was the default of our training code and was not tuned (Section 4.5). Metrics per round, on the exam set: pass@1 is the correctness of the *first* of the k samples (one draw per problem, so it carries sampling noise); pass@k; *sample accuracy*, the mean over all k samples; style counts over all generations (boxed, dfrac, dollar, ...); and per-problem added/lost counts between consecutive rounds, where a problem is *solved* in a round if at least one of its k samples is accepted. Added minus lost therefore equals $150 \times \Delta$pass@k, so these counts add only the gross churn behind the net change, not an independent test. pass@1 was fixed in advance; sample accuracy was not among the preset measures, but it estimates the same quantity from all k samples with less variance, so after the runs we chose it as the primary accuracy measure. Claims that rest on it, including the 2.2-point bound, are therefore post hoc in the choice of measure. "Correct" always means accepted by the verifier (preset `reward`), the same verifier that selects the training data.

Uncertainty. For every within-run change (round 0 against the last round) and every paired comparison of two runs on the same exam partition we report a 95% interval from a paired problem-level bootstrap: we resample the 150 exam problems with replacement (10,000 times), recompute the statistic for both rounds or runs on the same resampled problems, and take the 2.5th and 97.5th percentiles. Resampling problems rather than samples respects the clustering of the k samples within a problem. The per-sample correctness was recomputed from the saved generations with the same verifier; it reproduces the reported pass rates except for six round values that differ by one problem (0.6–0.7 points), most likely because of `math-verify` comparison timeouts (`experiments/viec3/bootstrap_ci.py`). These intervals cover the choice of problems and the sampling of generations for a given trained adapter; they do *not* cover training-seed variance, which we can only see informally across our two or three seeds, so they understate the uncertainty of run-level conclusions. Our own data show by how much: at 5×10^{-4} and seed 2, Runs A and B, whose round-0 training pools are identical (424 examples each), end 7.3 points apart in round-2 pass@1 (interval $[-14.0, -0.7]$, Section 4.5). The patch also changes how under 1% of exam generations are graded, and the later pools diverge (386 vs 398, 350 vs 332) as the two trajectories separate, but the starting difference

Table 2. Design P: pass@1 and pass@8 over five GVT rounds, pooled GSM8K + MATH-500, scored on the selection set. Round 0 is the untrained base model. Section 4.1 explains why these numbers cannot be attributed to any single cause.

Model	Metric	R0	R1	R2	R3	R4	Δ (R0→R4)
0.5B	pass@1	31.1%	31.9%	29.9%	28.9%	27.4%	−3.7 pt
0.5B	pass@8	60.5%	58.9%	56.1%	55.5%	54.9%	−5.6 pt
1.5B	pass@1	50.0%	46.2%	44.5%	32.4%	18.6%	**−31.4** pt
1.5B	pass@8	72.6%	71.5%	70.2%	63.6%	48.6%	**−24.0** pt

Table 3. Design P, 1.5B, split by dataset, recomputed from the saved generations (preset `reward`, the one used for selection; the `default` preset differs only at MATH-500 round-3 pass@8, 45.0%). Pooled, the recomputed round-3 pass@8 is 63.5% (`reward`) and 63.6% (`default`), against 63.6% in Table 2.

Dataset	Metric	R0	R1	R2	R3	R4	Δ
GSM8K	pass@1	69.6%	62.0%	61.0%	45.6%	26.6%	−43.0 pt
GSM8K	pass@8	91.4%	90.4%	89.0%	82.2%	65.8%	−25.6 pt
MATH-500	pass@1	30.4%	30.4%	28.0%	19.2%	10.6%	−19.8 pt
MATH-500	pass@8	53.8%	52.6%	51.4%	44.8%	31.4%	−22.4 pt

between the runs is almost nothing. An interval that excludes zero is therefore necessary, not sufficient, for a reproducible effect. We also make no correction for multiplicity: Table 7 alone holds 76 intervals, and at the 95% level about four would exclude zero by chance if nothing changed, so we rest conclusions on patterns across runs and seeds, not on any single interval. Single proportions use Wilson intervals.

4 Results

4.1 Design P: An Apparent Collapse

Table 2 reports pass@1 and pass@8 for both models over five rounds of the pooled design, as computed during the run (where the two verifier presets gave identical pooled values; a later per-dataset recomputation differs by one problem, see Table 3). The 1.5B decline is severe and accelerating; the 0.5B decline is small (−3.7 and −5.6 points). Splitting the 1.5B run by dataset (Table 3) shows both datasets falling heavily, although GSM8K has a measured 0% false-negative rate. The accepted pool also contracts every round (1.5B: from 3,962 to 1,566 solutions) and the 1.5B output style shifts, with `\boxed{}` falling from 5,608 to 2,693 of 8,000 generations while dollar-delimited math rises from 1,453 to 3,406.

Design P cannot be interpreted causally, for three reasons. First, each round trains one adapter on the union of both datasets' accepted solutions, of which GSM8K supplies 68.4–70.2% in every round, so either dataset's trajectory is a joint function of both pools. Second, it scores each round on the problems it

also used to select training data. We expected this to *raise* later-round scores, if anything. Design I lets us check that expectation, since it also generates on its 350 selection problems: there, sample accuracy on the selection problems falls as much as or more than on the exam (e.g. Run A at 5×10^{-4}: -8.2 and -5.3 points against -8.3 and -4.7 on the exam; at 10^{-4}, seed 1: -6.2 against -1.5). Training on a problem's own accepted solutions therefore did not protect it, and selection-set scoring is unlikely to have masked or created the pooled decline, though this is checked only in the other design. Third, and not anticipated when we designed the follow-up, Design P trains on far more examples per round: 3,962 at round 0 for the 1.5B model, against 872 for the isolated MATH-500 run at the same learning rate. With one epoch and the same batch size this is about 4.5× as many optimizer steps (Design P's training logs were not saved, so this is inferred from pool size). Section 4.5 suggests, on the evidence of one seed, that a higher learning rate produces more decline, with a falling `\boxed` share as here (pass@1 falls faster than pass@k in only two of its four runs at 5×10^{-4}), and Run C, which takes about twice Run A's steps per round at the same rate, is the one 10^{-5} run that declines clearly. One reading consistent with these data is that Design P's adapter changed the model more. It rests on two assumptions we have not tested: that more optimizer steps act like a higher learning rate, and that Run C's decline comes from its dose rather than its dataset (a single comparator). Pooling or scoring may also have contributed; no run separates these explanations.

4.2 Design I at 10^{-5}: Small Declines, a Larger One on GSM8K

Table 4 reports the isolated runs on their held-out exam sets. pass@8 ends within 4 points of its round-0 value in every run. On MATH-500, pass@1 is 2.7–5.3 points lower at round 1 than at round 0 and roughly flat afterwards, and 2.7–4.0 points lower at round 4; the bootstrap intervals for these round-0-to-round-4 changes are between $[-10.7, +2.7]$ and $[-9.3, +4.0]$, all including zero (Table 7). With one first-sample draw per problem per round, a step at round 1 cannot be distinguished from noise around a flat or slightly negative trend, so we claim only that there is no evidence of compounding on MATH-500 at this rate (Run C, below, loses most of its pass@1 in its last round), not that the first round has a one-off cost. The less noisy measures do move: sample accuracy falls by 1.2–3.2 points in these four runs, with one interval (B, seed 0) excluding zero, and the (post hoc) matched finished accuracy falls by 1.1–4.2 points, with intervals excluding zero at seed 0 for both A and B. These seed-0 intervals only just exclude zero (upper ends -0.6 and 0.0 for sample accuracy, -0.5 and -0.9 matched), A and B at seed 0 share their round-0 generations and almost all of their first pool, so they are close to one replicate, and at seed 1 nothing excludes zero. With about four of the 76 intervals in Table 7 expected to exclude zero by chance, we read this as a possible small decline at this rate, not an established one; what we do not see is a gain. Run C (GSM8K) declines more: pass@1 falls 12.7 points, 9.3 of them in the final round (63.3% → 54.0%; interval for the whole change $[-21.3, -4.0]$), and sample accuracy 13.1 points, while pass@8 falls 2.7

Table 4. Design I: pass@1 and pass@8 on the held-out exam set (150 problems), Qwen2.5-1.5B, learning rate 10^{-5}. Runs A and B differ only in the `\dfrac/\tfrac` patch. Each seed draws its own exam partition, so round-0 values differ between seeds.

Run	Metric	R0	R1	R2	R3	R4	Δ
A, seed 0 (MATH-500, orig.)	pass@1	32.0%	29.3%	28.0%	29.3%	29.3%	−2.7 pt
A, seed 0 (MATH-500, orig.)	pass@8	57.3%	56.7%	54.7%	57.3%	56.0%	−1.3 pt
A, seed 1 (MATH-500, orig.)	pass@1	36.7%	32.7%	32.7%	31.3%	33.3%	−3.3 pt
A, seed 1 (MATH-500, orig.)	pass@8	58.7%	60.7%	59.3%	59.3%	58.7%	0.0 pt
B, seed 0 (MATH-500, patched)	pass@1	32.0%	28.7%	28.7%	28.7%	28.7%	−3.3 pt
B, seed 0 (MATH-500, patched)	pass@8	57.3%	55.3%	52.0%	57.3%	53.3%	−4.0 pt
B, seed 1 (MATH-500, patched)	pass@1	37.3%	32.0%	32.7%	31.3%	33.3%	−4.0 pt
B, seed 1 (MATH-500, patched)	pass@8	59.3%	60.7%	58.0%	58.7%	58.7%	−0.7 pt
C, seed 0 (GSM8K, orig.)	pass@1	66.7%	66.7%	65.3%	63.3%	54.0%	−12.7 pt
C, seed 0 (GSM8K, orig.)	pass@8	92.7%	90.7%	93.3%	90.0%	90.0%	−2.7 pt

points (interval including zero) and the cap-hit share rises only from 3.5% to 4.2%. Finished accuracy falls 12.8 points matched, and the decomposition of Section 4.6 puts 97% of the sample-accuracy decline in finished accuracy, so this decline is not a termination effect. It is a single run, and it took about twice as many optimizer steps per round as Run A (951 against 436 at round 0), so we do not attribute it to the dataset alone.

How much training happened at 10^{-5}*.* The training loss does fall at this rate: within round 0 of Run A it goes from 0.72 to 0.38 (seed 0) and from 0.70 to 0.40 (seed 1) over about 420 steps, with an epoch mean of 0.47, against epoch means of 0.24–0.30 at 10^{-4}–5×10^{-4}. The adapter therefore learns, but slowly, and it changes the outputs little: mean length and the share of generations that hit the token cap are essentially unchanged from round 0 to round 4 (cap-hit $46.9 \to 47.4\%$ at seed 0 and $40.7 \to 41.8\%$ at seed 1 for Run A; Table 9). A flat pass@k curve here is evidence that this training step did little, not evidence that GVT is stable.

Noise. Round 0 is the untrained base model; only the exam partition and the sampling differ between seeds, yet round-0 pass@1 is 32.0% at seed 0 and 36.7% at seed 1, and on the third partition used later round-0 pass@4 is 40.0% against 50.0% and 51.3% on the other two (Table 6). Re-sampling the same model on the same problems moves pass@1 by up to 6.7 points (Section 4.6). A single 150-problem pass@1 has a standard error of about 3.8 points. We therefore no longer judge differences against one "noise floor"; we report the bootstrap intervals above and treat agreement across seeds as the only evidence about training-seed variance.

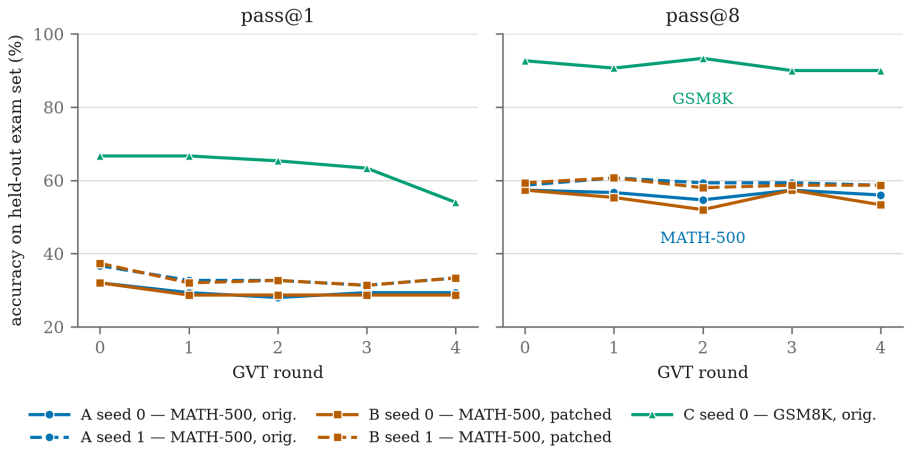


Fig. 1. Design I at 10^{-5}: pass@1 (left) and pass@8 (right) on the held-out exam set, by round. Colour identifies the run, line style the seed, marker the run again, so the figure survives greyscale printing. Runs A and B track each other closely at both seeds. pass@8 stays within a few points in every run. The one large visible movement is Run C's pass@1 in the final round.

4.3 The Patch Control Could Not Detect an Effect

Runs A and B are identical in dataset, seed, split, model and training configuration and differ only in whether `\dfrac`/`\tfrac` are normalised before verification. Their curves are close (Table 4): at round 4, B minus A is −0.7 [−3.3, +2.0] points of pass@1 at seed 0 and +0.0 [−2.7, +2.7] at seed 1 (95% intervals). This null is uninformative, for two reasons we should have checked before running it. First, the treatment is tiny: `\dfrac` appears in 20–29 of the 4,000 generations per round (0.50–0.73%), `\tfrac` never, and only the correct ones among those are wrongly discarded; the kept pools of A and B differ by 4 examples at seed 0 (872 vs 876) and 3 at seed 1 (833 vs 836), under 0.5% of the pool. Second, at 10^{-5} the adapter changes the outputs little (Section 4.2), so two nearly identical pools would give nearly identical models whatever the verifier did. We cannot separate these two explanations. The result is therefore: *removing one rare format defect had no detectable effect*, which the prevalence count alone predicts; it is not evidence about directional verifier bias in general, and it leaves untouched the gold-storage defect, which rejects 15.5–18.6% of correct in-loop solutions in both arms (Section 3.2). Verifier-bias figures in the literature, such as the ∼14% of [5], are likewise conditional on a reference distribution; what matters for a loop is how often the affected forms occur in that loop's own output, which is cheap to count.

Table 5. Runs A and B at 5×10^{-4} (MATH-500, $k = 4$, three rounds), held-out exam set of 150 problems. B differs from A only in the verifier patch; each seed draws its own exam partition. "Kept pool" is the number of training examples (= 2× optimizer steps). "Boxed" is the share of the 2,000 generations in each round containing `\boxed`.

Run, seed	Round	pass@1	pass@4	Kept pool	Boxed
A (orig.), 1	0	33.3%	50.0%	440	52%
	1	32.7%	49.3%	391	51%
	2	23.3%	42.7%	326	36%
B (patched), 1	0	33.3%	50.7%	443	52%
	1	35.3%	50.0%	392	49%
	2	28.7%	48.7%	328	41%
A (orig.), 2	0	31.3%	51.3%	424	54%
	1	31.3%	46.0%	386	50%
	2	27.3%	44.7%	350	37%
B (patched), 2	0	31.3%	52.0%	424	54%
	1	31.3%	44.7%	398	50%
	2	20.0%	40.7%	332	38%

4.4 Pre-Filter Answer Diversity

Over all problems of each run (in Design I the 500 selection and exam items together, not the exam alone) we also measured, before filtering, the fraction of problems on which all $k = 8$ samples give the same extracted answer. It falls in every run, more where pass@1 falls more: 24.8% → 1.7% for Design P's 1.5B run, 33.6% → 16.0% for Run C, 14.6% → 12.0% (A) and 14.6% → 12.2% (B) at seed 0. This agreement is mechanically tied to accuracy (problems on which every sample is correct are all-agreement problems) and to generations that hit the token cap, so it adds little beyond the accuracy change and does not by itself rule out stylistic convergence, which concerns how answers are written rather than which answer is given. The fraction of distinct answers among the k samples, which we had planned as the primary diversity measure, barely moves at 10^{-5} (0.542 → 0.554 for Run A, seed 0, the only run for which we computed it). We report both for completeness and draw no conclusion from them.

4.5 Higher Learning Rates

Because 10^{-5} changed the outputs little, we reran the MATH-500 runs at higher rates with a smaller budget (three rounds, $k = 4$, so pass@4 replaces pass@8 and the pools are about half as large): Runs A and B at 5×10^{-4} at two seeds, and Run A at 10^{-4} at three seeds and at 2×10^{-4} at two. Seed 1 shares the exam partition of the seed-1 runs above and seeds 2 and 3 draw new ones. These runs are not directly comparable with the $k = 8$, five-round runs in metric or training dose. Within round 0 the loss falls from about 0.5 to 0.25 at 5×10^{-4} and from about 0.6 to 0.28 at 10^{-4} over about 220 steps.

At 5×10^{-4} every run declines, with wide intervals. By round 2 pass@1 has fallen 10.0 and 4.0 points (A) and 4.7 and 11.3 points (B), and pass@4 7.3, 6.7, 2.0 and 11.3 points (Table 5). The bootstrap intervals (Table 7) exclude zero for two of the four pass@1 changes and one (two more reach zero at their upper end) of the four pass@4 changes, and for the sample accuracy of all four. A and B at the same seed share the partition, the round-0 generations and almost the whole first pool, so the four runs are closer to two independent replicates than four. We read this as a consistent but imprecisely measured decline. The style shift does not separate imitation from drift. The verifier-accepted pool is 89–91% `\boxed` at round 0, far more than all generations (52–54%, Table 5), because unfinished generations are rarely boxed and rarely accepted. But the right comparison is with *finished* generations, and these start *more* boxed than the pool (95.6% and 98.5% on the exam at seeds 1 and 2; the pool also holds truncated and cap-hit examples that lose their final `\boxed`). Among finished exam generations the share falls to 69.7% (A, seed 1) and 82.9% (A, seed 2), from above the round-0 pool to below it, while the pool, drawn from these generations, falls too (73–77% by round 2). A model imitating its less-boxed training data would move the same way, so this share cannot separate imitation of the pool from drift.

The decline may depend on the learning rate, on the evidence of one seed. Only seeds 1 and 2 were run at all three higher rates. At seed 1, sample accuracy at round 2 falls by 1.5, 3.5 and 8.3 points at 10^{-4}, 2×10^{-4} and 5×10^{-4}; at seed 2 by 3.7, 3.8 and 4.7, which is almost flat (Table 7; seed 3, run only at 10^{-4}, gives -1.0). The seed-1 series is monotone, the seed-2 series barely so, and at 10^{-4} and 2×10^{-4} the individual changes are about the size of re-sampling noise, so the rate dependence is suggestive, not established. The share of generations that hit the token cap does *not* follow the rate: it is flat or falls at 10^{-4} and 2×10^{-4} and rises only at 5×10^{-4} (Table 6).

The patch has no consistent effect. At round 2, B minus A is $+5.3$ [-2.0, $+12.0$] points of pass@1 at seed 1 and -7.3 [-14.0, -0.7] at seed 2. The sign flips with the seed, and the seed-2 interval excludes zero although the patch changes the first training pool by 0–3 examples. Same-seed runs that start from almost the same pool thus end several points apart once the adapter moves (their later pools differ by 1–18 examples as the trajectories separate), which is a direct look at trajectory noise in this regime.

Two controls at 5×10^{-4}. The *no-filter* run trains on all k samples of every selection problem regardless of correctness (1,400 examples per round), removing the verifier from the loop. The *train-on-gold* run ignores the model's generations and trains on the 350 reference solutions, identical every round. Both were run at two seeds, on the partitions of Run A at the same seeds. Neither is a single-variable control: per round, no-filter takes 700 optimizer steps and train-on-gold 175, against 220, 196 and 163 for Run A at seed 1 (and 212, 193, 175 at seed 2). Train-on-gold is thus close to Run A in optimizer steps, so its comparison is mainly one of data; no-filter takes 3–4× as many steps, and since dose may matter (the rate dependence above is suggestive), dose and data differ at once in that comparison.

Table 6. Run A (MATH-500, original verifier, $k = 4$, three rounds) across learning rates and seeds: pass@1, pass@4 and sample accuracy (600 exam generations) at round 0 $\rightarrow$ round 2, the share of round-2 generations that hit the 512-token cap (decoded text re-tokenised to ≥ 508 tokens; round 0 is 43.0–43.2%, 48.2% at seed 3), and problems added/lost summed over rounds 1–2. The two 5×10^{-4} rows are the A rows of Table 5.

Rate	Seed	pass@1	pass@4	Sample acc.	Cap-hit (R2)	Added/lost
10^{-4}	1	33.3 $\rightarrow$ 34.0	50.0 $\rightarrow$ 49.3	33.7 $\rightarrow$ 32.2	42.7%	20/21
10^{-4}	2	31.3 $\rightarrow$ 28.0	51.3 $\rightarrow$ 45.3	30.8 $\rightarrow$ 27.2	42.8%	25/34
10^{-4}	3	29.3 $\rightarrow$ 26.7	40.0 $\rightarrow$ 41.3	28.5 $\rightarrow$ 27.5	45.7%	16/14
2×10^{-4}	1	33.3 $\rightarrow$ 31.3	50.0 $\rightarrow$ 46.7	33.7 $\rightarrow$ 30.2	40.8%	23/28
2×10^{-4}	2	31.3 $\rightarrow$ 25.3	51.3 $\rightarrow$ 45.3	30.8 $\rightarrow$ 27.0	39.7%	21/30
5×10^{-4}	1	33.3 $\rightarrow$ 23.3	50.0 $\rightarrow$ 42.7	33.7 $\rightarrow$ 25.3	46.2%	22/33
5×10^{-4}	2	31.3 $\rightarrow$ 27.3	51.3 $\rightarrow$ 44.7	30.8 $\rightarrow$ 26.2	51.2%	26/36

Removing the filter has no effect of consistent sign. At round 2, no-filter minus Run A is +4.0 [−2.7, +10.7] (seed 1) and −4.7 [−12.0, +2.7] (seed 2) points of pass@1, of mixed sign, and −2.0 [−8.7, +4.7] and −5.3 [−12.0, +1.3] points of pass@4, negative at both seeds, so pass@4 weakly favours keeping the filter; for sample accuracy the difference is +0.2 [−3.3, +3.7] at seed 1 and −5.5 [−9.2, −1.8] at seed 2, so one single-seed interval excludes zero, in favour of the filter, and the other is centred on zero. The verifier gate keeps 30–31% of the samples at round 0 and 23–25% by round 2. With two seeds, a 3–4× larger training dose for no-filter, and intervals of this width, we can only say that effects of a few points in either direction are not excluded. Training on reference solutions is followed by a much larger decline: pass@1 falls to 18.0% and 15.3% (seed 1) and 8.7% and 10.0% (seed 2), and pass@4 by about 21 points by round 2 at both seeds ([−29.3, −12.7] at worst), while the model adopts the reference style (generations with a `$`-delimited span go from 5% to 98%, with `\boxed` from 52% to 86%). That accuracy is graded by the same verifier after a large format shift, so part of this drop could be verifier error. We checked this in two ways after the runs (post hoc). First, among the round-2 answers the verifier rejects, those whose final `\boxed` content equals the stored gold string after light normalisation are 14 of 600 generations at seed 1 and 11 of 600 at seed 2; counting them correct would raise round-2 sample accuracy by about 2 points. Second, the gold-storage re-score of Section 3.2 leaves the sample-accuracy decline at −24.7 and −22.0 points. Neither check is a full hand audit, but both put verifier error far below the size of the fall. Its rounds are replicate fine-tunes on identical data (their final training losses agree to three decimals), so the difference between its rounds 1 and 2 mainly reflects generation sampling. No verifier-filtered self-training run at this rate shows a pass@k gain; their pass@4 falls by 2.0–11.3 points (no-filter 9.3 and 12.0), whereas train-on-gold falls by about 21.

Run C at 5×10^{-4}. We also reran Run C (GSM8K, seed 1, three rounds, $k = 4$) at 5×10^{-4}, once. pass@1 falls 70.7% $\rightarrow$ 64.7% $\rightarrow$ 48.7% (interval for the round-

Table 7. Change from round 0 to the last round (percentage points) with 95% paired problem-level bootstrap intervals over the 150 exam problems (10,000 resamples; `experiments/viec3/bootstrap_ci.py`). Last round is R4 at 10^{-5} ($k = 8$) and R2 elsewhere ($k = 4$). "Finished, matched" is the last-round accuracy of generations below the 512-token cap minus the untrained model's accuracy on the same problem mix (Section 4.6; 2,000 resamples, `experiments/viec3/review_cpu/capanat.py`). The intervals do not include training-seed variance. Recomputed values match the reported ones except B, seed 1 at 10^{-5}, whose round-4 pass@8 is one problem lower (58.0% against 58.7%).

Run, seed	Δpass@1	Δpass@k	Δsample acc.	Finished, matched
Learning rate 10^{-5}				
A, 0	-2.7 $[-9.3, 4.0]$	-1.3 $[-8.0, 5.3]$	-2.4 $[-4.8, 0.0]$	-3.8 $[-7.2, -0.5]$
A, 1	-3.3 $[-10.0, 3.3]$	$+0.0$ $[-6.0, 6.0]$	-1.5 $[-4.1, 1.0]$	-2.3 $[-5.9, 1.2]$
B, 0	-3.3 $[-10.0, 3.3]$	-4.0 $[-10.7, 2.7]$	-3.2 $[-5.9, -0.6]$	-4.2 $[-7.8, -0.9]$
B, 1	-4.0 $[-10.7, 2.7]$	-1.3 $[-7.3, 4.7]$	-1.2 $[-3.7, 1.3]$	-1.1 $[-4.6, 2.6]$
C, 0	-12.7 $[-21.3, -4.0]$	-2.7 $[-7.3, 2.0]$	-13.1 $[-16.7, -9.8]$	-12.8 $[-16.5, -9.4]$
Learning rate 10^{-4}				
A, 1	$+0.7$ $[-6.7, 8.0]$	-0.7 $[-6.0, 4.7]$	-1.5 $[-5.3, 2.2]$	-4.3 $[-9.6, 1.1]$
A, 2	-3.3 $[-9.3, 2.7]$	-6.0 $[-12.7, 0.7]$	-3.7 $[-7.3, -0.2]$	-3.5 $[-9.2, 2.3]$
A, 3	-2.7 $[-10.0, 4.0]$	$+1.3$ $[-4.7, 7.3]$	-1.0 $[-4.3, 2.2]$	-0.6 $[-6.0, 4.7]$
Learning rate 2×10^{-4}				
A, 1	-2.0 $[-8.0, 4.0]$	-3.3 $[-10.0, 2.7]$	-3.5 $[-7.2, 0.0]$	-4.0 $[-8.9, 0.5]$
A, 2	-6.0 $[-12.7, 0.7]$	-6.0 $[-12.7, 0.7]$	-3.8 $[-7.5, -0.2]$	-4.7 $[-10.1, 1.0]$
Learning rate 5×10^{-4}				
A, 1	-10.0 $[-16.7, -3.3]$	-7.3 $[-14.0, 0.0]$	-8.3 $[-12.3, -4.5]$	-9.1 $[-14.2, -4.2]$
A, 2	-4.0 $[-11.3, 3.3]$	-6.7 $[-13.3, 0.0]$	-4.7 $[-8.2, -1.2]$	-5.5 $[-11.7, 0.1]$
B, 1	-4.7 $[-12.0, 2.7]$	-2.0 $[-8.7, 4.7]$	-6.7 $[-10.5, -2.8]$	-6.8 $[-12.4, -1.0]$
B, 2	-11.3 $[-17.3, -5.3]$	-11.3 $[-18.0, -4.7]$	-9.2 $[-12.8, -5.5]$	-5.9 $[-10.9, -1.0]$
No filter, 1	-6.0 $[-12.7, 1.3]$	-9.3 $[-16.7, -2.7]$	-8.2 $[-12.2, -4.3]$	-10.6 $[-16.4, -5.5]$
No filter, 2	-8.7 $[-16.0, -1.3]$	-12.0 $[-19.3, -4.7]$	-10.2 $[-14.2, -6.3]$	-6.6 $[-11.4, -2.0]$
Gold, 1	-18.0 $[-26.7, -10.0]$	-20.7 $[-28.7, -12.7]$	-19.7 $[-25.3, -14.2]$	-25.6 $[-32.1, -18.9]$
Gold, 2	-21.3 $[-28.7, -14.0]$	-20.7 $[-29.3, -12.7]$	-18.7 $[-23.7, -13.7]$	-25.9 $[-31.7, -20.0]$
C, 1	-22.0 $[-29.3, -15.3]$	-9.3 $[-15.3, -3.3]$	-15.3 $[-19.8, -11.2]$	-12.1 $[-16.6, -7.9]$

0-to-2 change $[-29.3, -15.3]$) and pass@4 86.0% → 86.0% → 76.7%. The cap-hit share rises from 3.8% to 15.0% and mean length from 958 to 1,313 characters, and accuracy on the finished generations falls clearly (71.9% → 63.3%; −12.1 points matched). In the three-part split of Section 4.6, about half of this run's sample-accuracy decline (51%) is in the finished-accuracy part and about half in termination, so on GSM8K, in this single run, the loss is both less accurate solutions and more unfinished ones. This run uses a different partition, k and round count from the 10^{-5} Run C, so the two can be compared only through their within-run changes (−1.3 points of pass@1 by round 2 at 10^{-5}, −22.0 at 5×10^{-4}; the 10^{-5} run falls further later, to −12.7 by round 4). That is suggestive of a rate dependence on GSM8K too, from one run per rate.

4.6 Anatomy of the Decline

Everything in this subsection is post hoc: the cap-hit measure, the matched metric and the decomposition were defined after the runs, to explain the declines, and they describe the declines rather than identify causes. Sample-level

Table 8. Controls at 5×10^{-4} (MATH-500, $k = 4$, three rounds), on the exam partition of Run A at the same seed. Pool is the number of training examples per round; optimizer steps are half of it.

Run	Pool	pass@1			pass@4		
		R0	R1	R2	R0	R1	R2
Seed 1							
A (verifier-filtered)	440→326	33.3	32.7	23.3	50.0	49.3	42.7
No filter	1400	33.3	26.7	27.3	50.0	49.3	40.7
Train on gold	350	33.3	18.0	15.3	50.0	33.3	29.3
Seed 2							
A (verifier-filtered)	424→350	31.3	31.3	27.3	51.3	46.0	44.7
No filter	1400	31.3	34.7	22.7	51.3	48.7	39.3
Train on gold	350	31.3	8.7	10.0	51.3	28.7	30.7

accuracy mixes two things: whether a model that finishes a solution gets it right, and whether it finishes within the 512-token cap at all. Token counts were not saved at generation time, so we re-tokenised the saved decoded text with the model's tokenizer and call a generation *cap-hit* if it has at least 508 tokens (`experiments/viec3/review_cpu/caphit.py`, `capanat.py`); the margin allows for small differences between generated and re-encoded token sequences. The threshold matters little: few generations fall near it (22–37 of 600 per run and round have 480–507 tokens), a threshold of 500 raises the cap-hit share by at most about 1.6 points and one of 512 lowers it by at most about 4.0, and with either the share still rises only at 5×10^{-4}. The untrained model already hits the cap in 43.0–43.2% of its MATH-500 exam generations (48.2% on the seed-3 partition, 3.5–3.8% on GSM8K), and only 4–7% of those are accepted. The 512-token cap is therefore binding from the start, not a consequence of training.

An earlier proxy overstated termination. Earlier versions of this paper called a generation *degenerate* if it was longer than 1,500 characters and contained no `\boxed`, and read its rise (9.8% → 41.3% for Run A, seed 1) as a growing failure to finish. Measured against cap-hit, the proxy flags only 9.8% of round-0 generations when 43.2% hit the cap, and its rise mostly tracks a change in the text rather than in termination: proxy-flagged generations have 3.3 characters per token at round 0 and 4.9–5.0 at round 2, against 2.7–3.0 for the rest, so drifting prose crosses 1,500 characters within the same token budget. At round 0 every proxy-flagged generation is a cap-hit (at round 2, 80–98% are), but the proxy catches only 19–25% of cap-hits at round 0 and 65–75% at round 2 (at 5×10^{-4}), so its rise is largely a rise in the share of cap-hits it can see. At 10^{-4} and 2×10^{-4} the proxy share grows 2.0–2.6× (at seed 1, 9.8% → 19.5% at 10^{-4} and 9.8% → 25.0% at 2×10^{-4}) while cap-hit stays flat or falls.

At the highest rate, finished solutions are less accurate. In all six self-training runs at 5×10^{-4} (Table 9) mean length grows by 41–52%, but the cap-hit share rises by only 2.3–14.2 points, while accuracy on the generations that finish falls

Table 9. What the decline consists of (MATH-500, exam generations, round 0 → last round). "Cap-hit" is the share of generations whose decoded text re-tokenises to ≥ 508 tokens; "finished acc." is sample accuracy on the other generations, unadjusted; "matched Δ" is last-round finished accuracy minus the untrained model's finished accuracy on the same problem mix (intervals in Table 7). "Trunc." is the percentage of the round-0 training pool longer than the 512-token training limit. The first six rows are at 5×10^{-4}, then 2×10^{-4} and 10^{-4}; the 10^{-5} rows are the original runs ($k = 8$, round 0 → round 4, not comparable in dose); the last two rows (train on gold) are at 5×10^{-4}.

Run	Seed	Length (chars)	Cap-hit (%)	Fin. acc. (%)	Matched Δ	Trunc.
A (orig.)	1	1175 → 1678	43.2 → 46.2	54.0 → 43.0	−9.1	24.5
B (patched)	1	1175 → 1659	43.2 → 47.8	54.0 → 46.3	−6.8	24.6
A (orig.)	2	1154 → 1641	43.0 → 51.2	50.6 → 45.4	−5.5	24.5
B (patched)	2	1154 → 1728	43.0 → 54.0	51.2 → 44.2	−5.9	24.5
No filter	1	1175 → 1668	43.2 → 45.5	54.0 → 43.1	−10.6	62.4
No filter	2	1154 → 1755	43.0 → 57.2	50.6 → 43.6	−6.6	—
A, 2×10^{-4}	1	1175 → 1433	43.2 → 40.8	54.0 → 46.5	−4.0	24.5
A, 2×10^{-4}	2	1154 → 1431	43.0 → 39.7	50.6 → 41.7	−4.7	24.5
A, 10^{-4}	1	1175 → 1276	43.2 → 42.7	54.0 → 48.5	−4.3	24.5
A, 10^{-4}	2	1154 → 1284	43.0 → 42.8	50.6 → 44.6	−3.5	24.5
A, 10^{-4}	3	1178 → 1355	48.2 → 45.7	50.2 → 47.5	−0.6	—
A, 10^{-5}	0	1158 → 1147	46.9 → 47.4	54.8 → 51.0	−3.8	—
A, 10^{-5}	1	1158 → 1146	40.7 → 41.8	55.2 → 54.3	−2.3	—
Train on gold	1	1175 → 485	43.2 → 7.2	54.0 → 14.7	−25.6	17.4
Train on gold	2	1154 → 516	43.0 → 8.3	50.6 → 12.5	−25.9	—

from 50.6–54.0% to 43.0–46.3%. Because the problems that still finish change, we compared last-round finished accuracy with the untrained model's finished accuracy on the *same problem mix*, weighting each problem's round-0 finished accuracy by the number of finished last-round samples it has. Problems with no finished last-round sample drop out (22–45 of the 150 per run); where a problem had no finished round-0 sample we use its accuracy over all round-0 samples (6–21 problems, 34–37 in the gold runs); intervals use 2,000 problem-level bootstrap resamples. Because training changes which problems finish, the matched metric conditions on an outcome of training and is descriptive, not a causal effect. On that matched basis finished generations lose 5.5–10.6 points, with intervals excluding zero in five of the six runs (Table 7). We also split the change in sample accuracy symmetrically into a finished-accuracy part $\bar{s}\,\Delta f$, a cap-share part $\Delta s\,(\bar{f} - \bar{c})$ and a cap-accuracy part $(1 - \bar{s})\,\Delta c$, where s is the share of finished generations, f and c the accuracy of finished and cap-hit generations, and bars denote round-0/last-round means. The finished-accuracy part is 73% of the decline for A seed 1, 62% for B seed 1, 74% for no-filter seed 1, but only 59%, 39% and 34% for A, B and no-filter at seed 2. Termination (the other two parts) is thus 26–41% in four runs and the larger part, 61% and 66%, for B and no-filter at seed 2. These six runs do not exhaust 5×10^{-4}: in the

replicate and the 1,024-token-limit run of Run A at seed 1 (below) the cap-hit share does not rise (43.2 → 43.0% and 43.8%), and the finished-accuracy part is 89% and 85% of the decline. At 10^{-4} and 2×10^{-4} the cap-hit share does not rise, the finished-accuracy part exceeds the whole (smaller) change in four of the five runs and is 93% of it in the fifth (10^{-4}, seed 2), and matched finished accuracy falls by 0.6 to 4.7 points with intervals including zero. Termination therefore contributes materially to the decline only at the highest rate, where its share varies from about a tenth to two thirds by run (a 10^{-5} run, seed 1, puts 63% of its 1.4-point decline in termination, but that decline is within noise). The long generations that remain open as ordinary solutions and then drift into loosely related prose (e.g. "... acknowledged hurdles acknowledged setbacks ..."), with low exact repetition (mean word 5-gram repeat ratio about 0.02, similar to other generations); this description rests on inspection of examples, not on a systematic annotation. Train-on-gold is different: responses shorten by 55–59%, cap-hit falls from 43% to 7–8%, and finished accuracy collapses (−25.6 and −25.9 points matched). This collapse may owe something to the training recipe rather than to the data alone: the loss is taken over the prompt as well as the response, and reference solutions are short and terse, so even one epoch on 350 of them per round can pull the model far from its own reasoning style; we did not test a response-only loss.

Two checks of the pipeline. The training limit (`max_length`= 512 tokens, prompt included) equals the generation cap, so 24–26% of the kept pool (108 of 440 examples in Run A at round 0) is truncated in training: it loses its final answer and end-of-sequence token and could teach the model not to stop. We reran Run A at seed 1 and 5×10^{-4} once with a training limit of 1,024 tokens, together with a matched replicate that keeps the 512-token limit. Removing the truncation changed little in this one run: the round-2 cap-hit share is 43.8% with the longer limit, 43.0% in the replicate and 46.2% in the original Run A, and matched finished accuracy falls by 5.7, 5.0 and 9.1 points. The differences between the three arms are within the spread we see between same-configuration replicates, and the longer limit does not remove every unterminated example: 43 of the 440 accepted round-0 examples (9.8%) are cap-hit generations, which have no end-of-sequence token under any training limit. With one run per arm this can only exclude a large truncation effect. The replicate reproduces the direction of Run A's decline (pass@1 33.3 → 31.3 → 25.3%, pass@4 50.0 → 48.0 → 44.0%).

Re-scoring the same adapters under a 1,536-token cap, with everything else fixed (Table 10), is consistent with the 512-token cap not being the whole story, though with one adapter per row and no intervals it does not establish it. The untrained model gains *more* from the larger cap than the adapters do (sample accuracy +6.9 points, against +5.9, +2.0 and +3.5), so the adapters' deficit does not close: −8.0, −8.8 and −7.8 points at 512 tokens and −9.0, −13.7 and −11.2 at 1,536, consistent with a deficit that lies mostly in what the adapters write rather than in where they are cut off. Since re-scoring the same adapter moves pass@1 by up to 6.7 points, differences of one to three points between rows are not interpretable. The data suggest that tripling the cap does not remove the

Table 10. Adapters re-scored on the same exam problems (seed-1 partition, $k = 4$, 150 problems) under a 512- and a 1,536-token generation cap. Round-2 adapters from runs at 5×10^{-4} ("replicate" keeps the 512-token training limit, "1024 limit" trains with no truncated example) and 2×10^{-4}, against the untrained model. One adapter per row; re-scoring the same adapter moves pass@1 by 2.7–6.7 points against the run's own round-2 value. Only summary statistics were saved for these re-scorings, so no bootstrap intervals and no cap-hit counts are available; "Degenerate" and "Acc. finished" use the character proxy of Section 4.6, which undercounts cap hits, and are shown only for completeness. Sample accuracy rests on 600 samples clustered in 150 problems.

Model	Cap	pass@1	pass@4	Sample acc.	Degenerate	Acc. finished
Untrained	512	36.0%	52.0%	36.3%	8.7%	39.2%
	1536	40.7%	59.3%	43.2%	2.5%	44.1%
5×10^{-4}, replicate	512	29.3%	46.7%	28.3%	30.0%	38.1%
	1536	35.3%	52.0%	34.2%	23.0%	43.3%
5×10^{-4}, 1024 limit	512	26.0%	45.3%	27.5%	37.2%	40.8%
	1536	30.7%	50.7%	29.5%	32.7%	41.3%
2×10^{-4}	512	27.3%	48.7%	28.5%	28.7%	37.9%
	1536	32.0%	52.7%	32.0%	23.7%	40.8%

gap, for single adapters on one partition; they do not show what a still larger cap would do.

5 Discussion

5.1 Answer to the Research Question

Does the model get better at mathematics? Not detectably. On held-out problems no run raises pass@1 or pass@k by an amount whose interval excludes zero, at any of four learning rates, and the sample-accuracy intervals exclude gains larger than about 2.2 points. This bound is loose on three counts: it is read off the observed intervals, not a pre-specified equivalence test; it uses sample accuracy, which we chose as primary after the runs (on the pre-set pass@1 and pass@k the upper ends reach +8.0 and +7.3); and the intervals omit training-seed variance, which is of the same size (a replicate of Run A at 5×10^{-4}, seed 1, ends 3.2 points of sample accuracy away). In every MATH-500 run except 10^{-4} seed 3 (16/14) the problems lost over the rounds equal or outnumber the problems gained (Table 6; 10^{-5} runs: 20/21 to 32/34 over four transitions), which restates the sign of the pass@k change rather than adding evidence. Most per-problem movement is churn: about 2 to 20 problems are gained and about as many lost per round even where the averages barely move.

Does it learn to write what the verifier accepts? The data cannot settle this. The pattern fixed in advance for this reading (pass@1 up, pass@k flat, style converging) requires pass@1 to rise, so it cannot fire in runs where accuracy falls, and pass@1 never rises detectably (the largest rise is +0.7 [−6.7, +8.0],

at 10^{-4}, seed 1); the rule is one-sided and its silence is not evidence against verifier-style learning. The post hoc `\boxed` share does not settle it either. The share of all generations with `\boxed` falls (52% → 36% and 54% → 37% at 5×10^{-4}, Table 5), but the kept pool (89–91% boxed at round 0) is less boxed than the *finished* exam generations it is compared with (95.6% and 98.5%), and the finished share falls (69.7% and 82.9%) as the pool's does (73–77% by round 2). That is what imitation of the pool would predict as well as what drift would, so the share does not discriminate. It is also not a clean measure of what the verifier prefers, because the gold-storage defect (Section 3.2) penalises some `\boxed` answers. The one strong style convergence (`$` spans 5% → 98%, `\boxed` 52% → 86%) happens under train-on-gold, with no verifier in the loop and with falling accuracy: style converging alone, which by our rule says nothing about verifier-style learning.

What does change? In this post hoc description, most clearly at the highest rate, the model gets less accurate on the solutions it does finish. The matched intervals exclude zero in five of six MATH-500 runs at 5×10^{-4}, in both GSM8K runs (10^{-5} and 5×10^{-4}) and, on MATH-500 at 10^{-5}, only just, for A and B at seed 0, which are close to one replicate; at 10^{-4} and 2×10^{-4} the point estimates are negative but the intervals include zero. More generations run into the token cap only at 5×10^{-4}, by 2–14 points in six runs (not in a replicate), and termination accounts for 26–41% of the decline in four of those runs and 61–66% in two (Section 4.6); at lower rates the cap-hit share changes by −3.3 to +1.1 points. The untrained model already hits the cap in 43% of MATH-500 generations, so the cap limits every round.

What happened to the pooled collapse? It does not reproduce when each dataset gets its own adapter and is scored on held-out problems. Because the pooled design also trained on about 4.5× more examples per round, and our higher-rate runs suggest (on one seed) that stronger training produces larger declines, we cannot say whether pooling, selection-set scoring or training dose produced it. We no longer claim that pooling and selection-set scoring "manufactured" the collapse.

5.2 Interpretation

An earlier version of this work explained the Design P curves with a self-reinforcing shrinking-pool mechanism: the biased verifier accepts a smaller, more style-skewed set each round and the skew compounds. The data give it no support, but they do not rule it out either. The patch removes a severe but rare bias and changes the pool by a handful of examples, so it tests how strong the bias is per answer, not whether a frequent bias compounds; the no-filter control declines about as much without any verifier, but it also trains on 3–4× more examples, so its decline is confounded with dose; and the training-dose difference between the designs offers a simpler account that we have not tested directly either. Pool size is also partly definitional (it is sample accuracy on the selection items times the number of samples), so its co-movement with accuracy, which

holds on pass@1 in Design I as well (Run C: pool −20%, pass@1 −12.7; Run A: −3%, −2.7), says little about mechanism.

Design I differs from Design P in at least four respects at once (isolation, a held-out split, n per run falling from 1,000 pooled problems to 500, and the training dose), so the P/I contrast is descriptive. The clean comparisons are internal to Design I and its higher-rate counterparts, and even there the controls differ in dose. The defensible statement about the verifier is narrow: *removing a format defect present in under* 1% *of generations, or removing the filter entirely, made no consistent difference at the resolution of this setup.* The one frequent systematic false negative we measured, the gold-storage defect, was present in every run and was never varied, so whether systematic false negatives are harmless in general, as Egashira et al. find for RLVR, our data cannot say.

5.3 Practical Suggestions

These suggestions come from one 1.5B model, one rule-based verifier, LoRA SFT and at most three seeds per condition; some are general good practice that our experience illustrates rather than findings. *Hold out evaluation items and keep one adapter per dataset* if you report per-dataset trajectories, so the curves can at least be read; our data do not rank these factors. *Match training dose across arms*, or report optimizer steps: pool size changes with filtering, k and dataset, and with one epoch it sets the number of updates. *Check that the training step changes the model* (loss and output statistics after one round) before reading a flat curve as stability. *Count the prevalence of a suspected verifier defect in the loop's own outputs before building a control around it;* a treatment that changes 0.5% of the pool cannot be detected with 150 problems. *Save token counts and report cap-hit directly,* not a character-length proxy, and adjust any "accuracy on finished solutions" for which problems still finish; our proxy misread a change in text style as a change in termination. *Report intervals,* and track pass@k and per-problem gains and losses alongside pass@1.

6 Limitations

1. *Unequal training doses.* With one epoch and batch size 2, optimizer steps are half the pool size: round 0 takes about 1,981 steps (inferred) in Design P, 436 in Run A and 951 in Run C at 10^{-5}, 220 in Run A at 5×10^{-4}, 700 for no-filter and 175 for train-on-gold. Every cross-arm comparison mixes the intended variable with dose, and the cause of the Design P decline is not identified.
2. *Underpowered controls.* The patch changes 0–4 training examples; at 10^{-5} the adapter also changes the outputs little. Its null is expected whatever the truth, and we report no equivalence bound. The no-filter and train-on-gold controls have two seeds each.
3. *Intervals cover only part of the uncertainty.* The bootstrap intervals cover problem draw and generation sampling, not training-seed variance; most

conditions have two or three seeds, Run C at the higher rate and the token-cap re-scorings one. pass@1 is a single draw per problem.

4. *Learning rate.* The original 10^{-5} was an untuned default; the evidence that it changes outputs little is behavioural (unchanged length and cap-hit share), since its loss does fall. The higher-rate grid is coarse and uses three rounds and $k = 4$.
5. *Cap-hit is reconstructed:* token counts were not saved, so we re-tokenise the decoded text and count a generation as a cap hit at ≥ 508 of 512 tokens. The token-cap re-scorings saved only summary statistics, so their termination figures still rest on the character proxy, which we show undercounts cap hits.
6. *Verifier measurements.* The false-negative audit used other models' solutions and a scripted correctness check, inspected only rejections (so no false-positive rate). The gold-storage defect was measured afterwards on the loop's own outputs by re-scoring (Section 3.2); it lowers all our absolute pass rates by about 5–8 points for self-training, was present in every run including both arms of the patch control, and was never varied in a training run. The re-scored false-negative share assumes the re-scored verifier makes no false accepts.
7. *Restart-from-base design and scope.* A fresh adapter on the base model each round leaves compounding little room, so the absence of compounding is partly a property of the design. One model size in Design I, one verifier, one training method, at most five rounds; full fine-tuning, continued training and online RL are untested.
8. *The selecting verifier also grades.* The same math-verify configuration that chooses the training data scores the exam, so any systematic verifier error enters both selection and measurement. The gold-storage re-score (Section 3.2) and the string audit of the train-on-gold runs bound part of this, but only for five runs and one defect class.
9. *A binding token cap.* At 512 tokens the untrained model already fails to finish 43% of MATH-500 generations, so every measure is taken in a regime where length and accuracy are entangled, and the 512-token training limit truncates about a quarter of the kept pool. Our conclusions are about this regime; a larger cap during the loop is untested.
10. *No equivalence margin and many intervals.* We did not fix in advance how small a gain would count as "no gain", so the 2.2-point bound is read off the observed intervals. We report 76 intervals without a multiplicity correction; at the 95% level about four would exclude zero by chance alone, so single intervals that only just exclude zero should be read cautiously.
11. *Possible test-set contamination.* MATH-500 and GSM8K are public and may be in Qwen2.5's training data. This would raise absolute accuracy but should affect all arms alike; we did not test it.
12. *Training recipe.* The loss covers the prompt as well as the response, and we tried one LoRA configuration (rank 16 on query and value projections). Both could matter, most plausibly for the train-on-gold collapse, and neither was varied.

7 Conclusion

We asked whether generate–verify–select–train iterations make a 1.5B model better at mathematics or only teach it to write what a rule-based verifier accepts. On a held-out exam, the first answer is no: on the four measures fixed in advance no run improves detectably, and the intervals for sample accuracy (a primary measure chosen after the runs) exclude gains above about 2.2 points, a loose bound because they omit training-seed variance and are not an equivalence test. The second stays open, because our pre-set rule for it could only fire if accuracy rose, and the post hoc `\boxed` share falls in a way that imitation of the less-boxed training pool and drift would both produce. In a post hoc description, most clearly at the highest rate, the model becomes less accurate on the solutions it does finish; more generations hit the token cap only at the highest rate, where termination accounts for a quarter to two thirds of the decline depending on the run, and a character-length heuristic we first used overstated that part. Reference-solution training degrades the model more, with much shorter answers. Patching a rare verifier defect and removing the filter made no difference of consistent sign across measures and seeds (pass@1 flips sign between seeds; removing the filter lowers pass@4 at both seeds, within noise; single-seed intervals sometimes exclude zero), neither control could have detected a small effect, and a frequent defect in our gold storage, which rejects about one in six correct solutions, ran unvaried through every experiment. A sharp decline seen in an earlier pooled design does not reproduce under isolation and held-out scoring, and because that design also trained several times longer per round, we leave its cause open. We report the earlier readings that did not hold up because the designs that produced them are common.

Future work. Separate the explanations of the pooled decline with a pooled run scored on a held-out split and an isolated run matched in optimizer steps; save token counts at generation time and test a larger generation cap during the loop, since the untrained model already hits 512 tokens in 43% of MATH-500 generations; refine the learning-rate grid and replicate Run C; and run a positive control (a verifier biased against a frequent format) to show what effect size this design can detect.

The gold-storage defect (Section 3.2) is the natural next verifier control. We implemented and unit-tested a patch that wraps the gold in `\boxed{}` whenever the prediction contains one and the gold does not; re-scoring shows that in a training run it would add about 17–23% to the kept pool (measured against the size of the round-0 kept pool), roughly 35–50 times the change made by the `\dfrac` patch, so unlike Run B such a run would actually perturb the training data. Given that same-configuration runs end several points apart, a single seed per arm would still detect only a large effect.

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